DinHot: A Holographic Optical Trap Studio Comparing Classical, State-of-the-Art and Beyond-State-of-the-Art Hologram Methods on a Device Model

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github.com/fsantibanezleal/CEFOP_DinHot

Abstract

A phase-only spatial light modulator in the pupil of a microscope objective can form hundreds to thousands of optical traps at once, but only if the phase pattern it displays is computed well: the pattern fixes how much light reaches the traps, how equal the traps are, where stray light goes, how the pattern survives a real device, and how fast it can be recomputed when traps move. DinHot is an open studio that runs 24 ladder methods (10 classical, 9 state of the art, 5 beyond it) on the same task, with the same seeds, metrics and device model, in Python as the reference and in a TypeScript port that runs live in a browser. We describe the optical and device model, the methods with the measured acceptance of each, 13 cases from a 3×3 array to a disordered 819-trap array and shaped potentials, the benchmark protocol, two learned models, and the 5 proposals past the published state of the art, three of which meet their criteria (a device-aware weighted Gerchberg–Saxton update, a flicker-bounded phase-continuous sequence and an efficiency–uniformity front), one of which meets a narrowed criterion (an aberration-robust hologram) and one of which is refuted by measurement (a learned warm start for phase-fixed weighted GS). A network trained on three-plane volumes beats its baseline at equal wall-clock on all 162 held-out volumes (median accuracy ratio 1.140), whereas its two-dimensional counterpart ties with Gerchberg–Saxton (ratio 0.9994). The model is a scalar Fourier model of a simulated device, not a calibrated instrument.

Keywords: holographic optical tweezers, spatial light modulator, weighted Gerchberg–Saxton, pixel crosstalk, neutral-atom arrays, hologram benchmarking, learned holography.

1 Introduction and problem

Holographic optical tweezers shape one laser beam into many traps with a phase-only spatial light modulator (SLM). The largest published neutral-atom tweezer array holds 6,100 atoms in about 12,000 tweezers, with a trap-depth spread of 11.4 percent across its sites [1]. The spread matters: sites with unequal depth differ in loading and in gate fidelity, and the light that forms them is hard to keep uniform as the array grows. The same computation serves optical micromanipulation of beads and cells, neuron stimulation and laser machining, with different limits on power, speed and the number of traps.

The hologram methods that solve this problem span half a century: the superposition of prisms and lenses, random superposition and random masks, the Gerchberg–Saxton (GS) iteration and its adaptive and weighted forms [2], direct search [5], conjugate-gradient minimization [6], phase-fixed weighted GS for large arrays [3], mixed-region amplitude freedom for shaped potentials [4], gradient descent through differentiable forward models [7], optimal-transport initialization [15, 16], and learned holography [14]. Camera feedback closes the loop on a real set-up [8]. A search of the literature found no controlled published comparison of weighted GS,

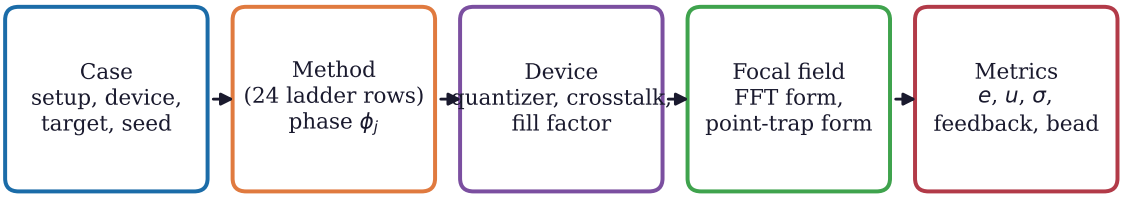


Figure 1: The chain every case runs: a case (optical setup, device, target and seed) is given to a method, which returns a phase; the device model turns the phase into the field the instrument forms; the metrics score the traps, and the feedback and particle simulations act on the scored field.

phase-fixed weighted GS, conjugate gradient, MRAF and a learned method on the same point-trap array, and no public tool that runs them side by side with a device model and the physics of what is trapped.

DinHot provides that comparison. It began at CEFOP, Universidad de Concepción, as DinHotSys, a C++ desktop tool for dynamic holographic optical tweezers dated 2010; it was rewritten in Python and FastAPI and is rebuilt here as a static browser application whose numbers are computed by the engine or baked offline and replayed. This report describes the problem, the optical model and the device, the three tiers of methods, the cases and the protocols, the results, the learned models, the proposals beyond the published state of the art, the live engine and the limits. Every number in the tables and the text is generated from the committed artifacts of the repository or is quoted from a named unit measurement.

2 The optical model and the device

2.1 Forward models

A phase-only SLM of $n_{\mathrm{pix}} \times n_{\mathrm{pix}}$ pixels with pitch p sits in the pupil plane of a lens of focal length f and wavelength λ. For a trap m at (x_m, y_m, z_m) with vortex charge l_m the phase delay at pixel j is

$$\Delta_j^m = \frac{\pi z_m}{\lambda f^2}(x_j^2 + y_j^2) + \frac{2\pi}{\lambda f}(x_j x_m + y_j y_m) + l_m \,\mathrm{atan2}(y_j, x_j), \tag{1}$$

and the trap field and intensity are

$$V_m = \frac{1}{N_{\mathrm{pix}}} \sum_j A_j e^{\,i(\phi_j - \Delta_j^m)}, \qquad I_m = |V_m|^2, \tag{2}$$

with A_j the input beam amplitude normalised so that $\sum_j A_j^2 = N_{\mathrm{pix}} = n_{\mathrm{pix}}^2$, following Di Leonardo et al. [2]. I_m is the fraction of the input power in the peak of trap m. The second form is the centred, zero-padded, orthonormal fast Fourier transform (FFT) of the SLM field; at a trap on an FFT bin with $z = 0$ and no vortex the two forms agree exactly, which the test suite checks to 10^{-9} rad. One bin of the padded transform spans $\lambda f/(\mathrm{pad}\, n_{\mathrm{pix}}\, p)$ and the unaliased half-window is $\lambda f/(2p)$; a trap outside it is refused with the bound (case C11). Depth is handled by the angular spectrum with the band limit of evanescent components removed; the model is scalar and paraxial, so a high numerical aperture is flagged as outside its validity (case C07).

2.2 Metrics

With I_m the trap intensities divided by their target weights, DinHot reports

$$e = \sum_m I_m, \qquad u = 1 - \frac{\max I - \min I}{\max I + \min I}, \qquad \sigma = 100\,\frac{\mathrm{std}(I)}{\mathrm{mean}(I)}\ \%, \tag{3}$$

the efficiency, the uniformity ($u = 1$ is perfect) and the relative standard deviation. Polin et al. [5] use the same ratio without the leading $1-$, a non-uniformity; the two are not mixed. A power budget separates the traps, the zero order, the ghost traps (everything else on the grid, including folded replicas) and the remainder.

Table 1: The optical setups of the cases, read from the committed manifests and traces. The device row of C08 is the device the benchmark also uses for its device cells.

Setup	λ (nm)	f (mm)	NA	SLM pixels	pitch (μm)	pad	bin (μm)	half field (μm)
Atom (C01–C04, C09–C13)	850	6	0.5	256^2	20	2	0.498	127.5
Bead (C05–C07, in water, index 1.33)	1064	2.4	1.2	256^2	10	2	0.499	127.7
C08	850	3	0.5	128^2	20	2	0.498	63.8

C08 device: 4-bit phase, crosstalk $\sigma = 1.24$, $q = 1.80$ on 3×3 sub-pixels, fill factor 0.85, flicker 0.05 rad.

2.3 The device

A real SLM quantizes the phase, couples neighbouring pixels, covers only part of each pixel and fluctuates. The device model applies, in order, wrapping, quantization to 2^b levels, a nonlinear response, Gaussian flicker of stated rms and wrapping, and blurs the upsampled phase with the crosstalk kernel

$$K = \mathcal{F}^{-1}\Big\{ \exp\Big[-\frac{|k_x|^q + |k_y|^q}{\sigma^q} \Big] \Big\}, \qquad \sigma = 1.24,\ q = 1.80, \tag{4}$$

the fitted parameters of Schroff et al. [9] on 3×3 sub-pixels (error below 2 percent in that work). A square pixel of fill factor fill multiplies the focal field by a product of cardinal sines, so the intensity falls to about 0.4 at the edge of the unaliased window; this envelope is derived in the engine and tested against a brute-force transform. The adjoint of the crosstalk operator is closed form (convolution with the reversed kernel and a block sum), which lets gradient methods differentiate through the device without automatic differentiation. Zernike aberrations follow the Noll indexing [10]. Table 1 lists the three optical setups of the cases.

3 The three tiers of methods

The design document fixes, before any code, a ladder of 24 rows: 10 classical methods (the established methods of the problem), 9 state-of-the-art methods (current solutions, replicated or implemented, never only cited) and 5 proposals beyond the state of the art. Each row has one measurable acceptance criterion and one test. Table 2 lists them; Table 3 gives what each test measured on DinHot's own grids, with the seeds stated.

3.1 Classical tier

Analytic baselines (M01 to M03). The superposition of prisms and lenses, random superposition (each trap's phase drawn at random) and the random mask (each pixel assigned to one trap) are the references for the other methods. On the calibration task of Di Leonardo et al. [2] they show the published order: superposition is efficient only by accident and nearly non-uniform, random superposition gains efficiency at a spread, and the random mask has an efficiency of about the inverse of the number of traps. **Gerchberg–Saxton family (M04 to M06).** GS iterates between the SLM and the trap fields; the generalized adaptive additive form (GAA) adapts the update; weighted GS (GSW) rescales the target weights by the ratio of the mean to the achieved trap amplitude, $w_m \leftarrow w_m \langle r \rangle / r_m$, and reaches uniformity close to one [2]. **Direct search (M09)** minimizes the cost $C = -\langle I \rangle + f\sigma$ pixel by pixel over eight phase levels [5]. **Zernike correction (M14)** recovers injected aberrations by a sensorless modal search [10]. **Laguerre–Gaussian vortex traps (M15)** wind the ring phase by $2\pi l$. **Multi-plane GS (M19)** propagates plane by plane with an amplitude constraint per plane [11].

3.2 State-of-the-art tier

Phase-fixed weighted GS (M07) freezes the focal-plane phase after N iterations so that later iterations adjust only amplitudes [3]; the source reports 1500 foci at 91.2 percent modulation efficiency and camera correction to 1.59 percent in three steps against 6.47 percent for conventional weighted GS. DinHot finds that the source default of $N = 12$ stalls on a crowded 50×30 array and uses $N = 60$ there (Table 3). **MRAF (M08)** frees a region outside the signal and trades efficiency for accuracy on shaped potentials [4]. **Conjugate-gradient minimization (M10)** uses powered intensity costs and an analytic gradient obtained from two FFTs [6]. **Wirtinger gradient descent with Adam (M11)** differentiates any forward model

through the complex gradient [7]. **Closed-loop weighted GS (M12)** corrects trap amplitudes from a camera; DinHot runs it on an emulated camera whose errors are stated parameters, after Nogrette et al. [8]; the emulator reproduces the published feedback curves in the browser within the factor set in the design document (Table 3) and serves as a validation tool, not as a proposal. **Device-aware conjugate gradient (M13)** places the crosstalk kernel in the loop [9]. **Hungarian assignment and linear phase interpolation (M16)** sorts atoms with a cost $\sum d^{\alpha}$, $\alpha > 1$, which avoids crossings, and moves them along interpolated holograms [12, 13]. **An unsupervised convolutional network refined by GS (M17)** follows DeepCGH [14]. **Optimal-transport initialization of MRAF (M18)** seeds the phase from a transport map [15, 16].

Of the 24 engine methods, 22 ran in the bake of the cases; B3 and M17 are the learned slots, whose baselines answer when no model is released (Section 6).

Table 2: The method ladder of the design document: identifier, tier, method and acceptance criterion.

Id	Tier	Method	Acceptance criterion
M01	Classical	Superposition of prisms and lenses	on the 10x10 task with a flat beam, efficiency within 0.10 of 0.29 and uniformity below 0.05
M02	Classical	Random superposition	on the 10x10 task with a flat beam, efficiency above M01 and at least 0.65; uniformity below 0.35 (DinHot measured 0.25 at 256 squared against 0.01 at 768 squared in the source; the ordering, not the value, transfers)
M03	Classical	Random mask	efficiency about 1/M; at the source's 768 squared grid, uniformity within the spread of 0.58
M04	Classical	Gerchberg-Saxton, point-trap and FFT forms	the two forms agree to 10^{-9} rad; 30 iterations on the 10x10 task with a flat beam reach efficiency at least 0.9 and uniformity near 0.6
M05	Classical	Generalized adaptive additive	uniformity above GS at equal iterations on the 10x10 task
M06	Classical	Weighted GS	efficiency and uniformity at least 0.9 within a median of 15 iterations on the 10x10 task with a flat beam (the source reports about 10); the 3x3x3 lattice above 0.9 and 0.98
M07	State of the art	Phase-fixed weighted GS	fewer iterations than M06 to a relative standard deviation of 0.5 percent on a 50x30 array, with the phase fixed after 60 iterations (DinHot's measured default; at the source's 12 it stalls at 2.9 percent on this grid)
M08	State of the art	MRAF	on disc, ring and bar targets, signal-region error at least five times below GS at the cost of efficiency
M09	Classical	Direct search	uniformity at least 0.98 on the 10x10 task; under 5 percent rms variation on a quasi-periodic set
M10	State of the art	Conjugate-gradient minimization	the analytic gradient matches finite differences; a square lattice below 1 percent error
M11	State of the art	Wirtinger gradient descent with Adam	lower trap-intensity error than M04 at equal iterations on the 10x10 task
M12	State of the art	Closed-loop WGS on an emulated camera	from a 19 percent rms open-loop spread, at most 2.1 percent in 20 iterations at gain 0.7
M13	State of the art	Device-aware CG with pixel crosstalk	on the crosstalk device, lower error than M10 without the model on a large disc
M14	Classical	Zernike correction and sensorless modal recovery	recovers an injected set of Noll modes 4 to 11 within 5 percent of their amplitude
M15	Classical	Laguerre-Gaussian vortex traps	the ring phase winds by 2 pi l with an error under 0.1 rad
M16	State of the art	Hungarian assignment and linear phase interpolation	no crossing moves for alpha 1.5 on a 7x7 to 3x3 sort; a moving trap above 70 percent of its mean intensity
M17	State of the art	Unsupervised CNN hologram refined by GS (DeepCGH-style start)	on held-out targets, the network start followed by GS refinement inside the same wall-clock budget reaches a higher accuracy than GS alone (measured alone, the network matches GS at equal time: median ratio 0.998)
M18	State of the art	Optimal-transport initialized MRAF	lower error than MRAF from a random start on the shaped targets

Id	Tier	Method	Acceptance criterion
M19	Classical	Multi-plane GS	on a three-plane target, every plane mean within 25 percent of the overall mean (volumetric 3D GS of Whyte and Courtial)
BS-1	Beyond	Device-aware weighted GS (engine id B1)	on the crosstalk device, relative standard deviation at least two times below M07, efficiency at most 10 percent lower
BS-2	Beyond	Flicker-bounded phase-continuous sequence (engine id B2)	minimum transient intensity of a moving trap at least 0.9 of its mean, against LPI
BS-3	Beyond	Learned warm start for phase-fixed WGS (refuted) (engine id B3)	measured negative: on held-out geometries the network start needs a median 8 iterations to 1 percent against 14 from a random start (1.75 times, below the 2 times claimed), and a control with the same early phase fix and no network gains as much; the model is not released and the claim BS-3 is reported as refuted
BS-4	Beyond	Aberration-robust hologram (engine id B4)	over a 200-sample ensemble of residual aberrations, the 95th percentile of the relative standard deviation at least 1.25 times below M07, efficiency at most 10 percent lower; robustness to pointing errors is measured and reported, not claimed (limited by the spot width: 1.02 times)
BS-5	Beyond	Efficiency-uniformity Pareto front (engine id B5)	hypervolume at least that of the M04, M05, M09 and M06 points on the 10x10 task

Table 3: Measured acceptance of the methods on DinHot's own grids (unit tests; seeds as stated). Intervals are [min, max].

Id	Measured	Seeds
M01	efficiency 0.281, uniformity 0.0003 (deterministic)	1
M02	efficiency 0.750 [0.738, 0.752], uniformity 0.252 [0.106, 0.272]	6
M03	256 squared: efficiency 0.010, uniformity 0.057 [0.031, 0.072]; 768 squared: efficiency 0.008, uniformity 0.566 [0.543, 0.590]	5 and 2
M04	flat beam: efficiency 0.939 [0.933, 0.939], uniformity 0.604 [0.557, 0.626]; Gaussian beam: efficiency 0.815	5
M05	uniformity 0.688 [0.638, 0.709] against 0.615 [0.550, 0.637] for GS at equal iterations	5
M06	efficiency and uniformity both at 0.9 after 10 to 13 iterations (flat and Gaussian beams); 3 x 3 x 3 lattice reaches 0.9 and 0.98	3
M07	50 x 30 array to 0.5 percent in 63 and 64 iterations against 121 and 147 for M06 (phase fixed after 60); with the source value of 12 it stalls at 2.88 percent	2
M08	signal-region error 14.5 (disc), 7.9 (ring) and 12.1 (bar) times below GS; efficiency 0.18 against 0.97 to 0.99	3
M09	uniformity 0.9994 and 0.9992 (efficiency 0.66 and 0.65); 119 quasi-periodic traps at 0.023 percent rms	2 and 1
M10	analytic gradient within 1.8e-7 of finite differences; square lattice error 0.55 and 0.51 percent	2
M11	relative standard deviation 2.54, 2.46 and 2.26 percent against 17.79, 17.98 and 18.82 for GS at 30 iterations	3
M12	emulated camera loop: open-loop 20.9 and 20.1 percent to 1.47 and 1.08 percent in 20 iterations; 24.3 and 23.4 percent to 0.77 and 0.53 percent in three phase-fixed steps	2
M13	error 0.241 (efficiency 0.535) against 1.38 (0.507) for M10 without the device model, large disc on the crosstalk device	1
M14	worst per-mode relative error 0.002 [0.001, 0.002] over Noll modes 4 to 11	3
M15	ring phase winds by 2 pi l with a maximum error of 0.033 rad [0.028, 0.060] (isolated vortex)	3
M16	0 path crossings in 60 loadings; moving traps keep 0.870 [0.748, 0.963] of their mean intensity	60 loadings
M17	three-plane network refined by M19: median accuracy ratio 1.140 [1.102, 1.160] on 162 held-out volumes	162 volumes
M18	signal-region error 0.0204 (disc), 0.0179 (ring) and 0.0221 (line) against 0.250, 0.192 and 0.250 for MRAF from a random start	1
M19	plane means over overall mean 0.938, 0.987, 1.075 and 0.926, 1.034, 1.039 on three planes	2
BS-4	95th percentile of the relative standard deviation 4.045 against 6.268 for M07 (1.55 times) over a 200-sample aberration ensemble; efficiency 0.7031 against 0.7169; pointing errors 1.02 times	1

Table 4: The thirteen cases: category, optical setup, target and the methods that ran in the bake. C11 is refused by the input contract and never computed.

Case	Category	Setup	Target	Methods	Reference
C01	Arrays	850 nm, NA 0.5	9 traps	13	M06
C02	Arrays	850 nm, NA 0.5	100 traps	13	M07
C03	Arrays	850 nm, NA 0.5	1024 traps	6	M07
C04	Shaped and 3D	850 nm, NA 0.5	disc potential, 8-bit	6	M13
C05	Shaped and 3D	1064 nm, NA 1.2	51 traps	10	M09
C06	Shaped and 3D	1064 nm, NA 1.2	25 traps	8	M07
C07	Device and optics	1064 nm, NA 1.2	25 traps	12	M07
C08	Device and optics	850 nm, NA 0.5	64 traps, 4-bit, fill 0.85	7	B1
C09	Device and optics	850 nm, NA 0.5	16 traps, aberrated	6	M14
C10	Dynamics and controls	850 nm, NA 0.5	100 traps	4	M16
C11	Dynamics and controls		one trap at 200 μm, outside the field	0	refused
C12	Arrays	850 nm, NA 0.5	100 traps	7	M06
C13	Arrays	850 nm, NA 0.5	819 traps	6	M07

Id	Measured	Seeds
BS-5	hypervolume 0.8201 against 0.8033 for the union of the GS-family points; one front point dominates M04 and M05	1

4 The cases

A case is an optical setup, a device, a target (a list of traps or a shaped potential), an aberration, a method set, a seed and, for dynamic cases, a particle and a rearrangement. There are 13 cases in four categories (Table 4): arrays (a 3×3 grid, a 10×10 grid with camera feedback, a 32×32 array of 1024 traps, the calibration task and a disordered array of 819 traps on integer FFT bins at least five bins apart), shaped and three-dimensional targets (a disc potential with the crosstalk model, a three-plane array with vortices, a vortex ring with a line), device and optics (high numerical aperture, a device-limited array, an aberrated array with and without correction) and dynamics and controls (an atom-rearrangement sequence and a trap outside the field, which is refused and never computed). Every case states numeric expected ranges for its metrics, measured on three seeds, and the bake fails outside them. All cases are synthetic: the parameters are stated and no measured sample is used. The baked cases span 9 to 1024 traps.

5 Protocols and results

5.1 Baked cases

Each case is computed offline with every method valid for it (the seed is stated in each manifest), checked against its expected ranges, and committed as a manifest and a trace of at most 256 KiB. Table 5 compares the calibration case with the published rows of Di Leonardo et al.: the ordering of the six baselines and the weighted-GS uniformity are reproduced; the efficiencies of the GS family are lower than published because the engine's Gaussian input beam caps the efficiency, whereas the source uses a flat beam (the flat-beam test of M04 gives 0.939 efficiency and 0.604 uniformity; Table 3). The DinHot uniformity of random superposition and of the random mask differ from the published values on the engine's smaller grid, as the acceptance test of those methods states: the order, not the value, transfers.

Table 6 summarizes the baked cases. Where both ran, phase-fixed weighted GS (M07) leaves a median relative standard deviation of 0.27 percent against 18.7 percent for plain GS (M04) over 9 cases, at efficiencies between 0.453 and 0.817. On the device-limited case C08 phase-fixed weighted GS (M07), which ignores the device, keeps a spread of 14.8 percent, and the device-aware weighted GS (B1) lowers it to 6.7 percent, a factor of 2.2, on that small grid (the larger task of Section 7 gives a larger factor; both are reported in the companion manuscript). On the aberrated case C09, correction of the Zernike modes (M14) restores efficiency and uniformity that the uncorrected methods lose.

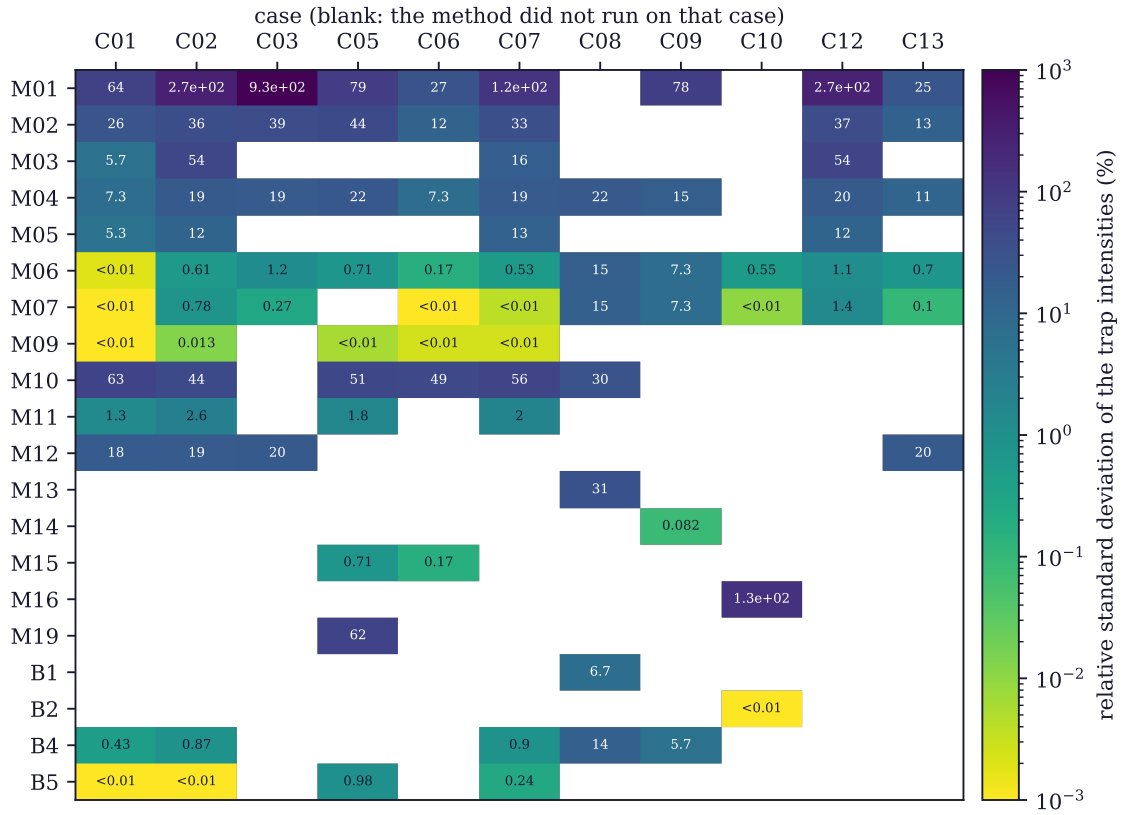


Figure 2: Relative standard deviation of the trap intensities, every method on every point-trap case of the bake (the shaped-potential case C04 is scored on its signal region and is omitted). Each cell is one method run at the case budget. Plain Gerchberg–Saxton (M04) leaves the largest spreads; the weighted and phase-fixed forms (M06, M07), the direct search (M09), the Pareto front (B5) and the device-aware methods remove most of it where they apply.

5.2 Feedback and particles

The emulated camera loop reduces the spread as in the published curves. In the baked traces the relative standard deviation falls from 9.1 to 0.02 percent in 5 steps on C01, from 20.3 to 0.21 percent in 20 steps on the 10×10 grid C02, from 22.3 to 1.36 percent in 3 steps on the large array C03, and from 22.0 to 0.52 percent in 5 steps on the disordered array C13 (Fig. 4). The published closed-loop result of Nogrette et al. [8] is a trap-depth spread of 19 percent rms reduced to 1.4 percent in 20 iterations; the unit test of M12 reproduces it within the factor set in the design document (Table 3). The particle simulations act on the computed traps: C01 carries a bead in the central trap (stiffness 4.01 pN/μm and a corner frequency of 70.0 Hz in the trace) and C03 an atom (a corner frequency of 104800 Hz and a radial trap frequency of 107292 Hz in the trace); the trap depth follows the intensity [19], and the bead's spectrum is a Lorentzian whose corner frequency is the stiffness over $2\pi\gamma$ [20], which the particle oracle checks.

5.3 The benchmark protocol

The benchmark compares the tiers on one protocol. Its geometries are (7) a 10×10 square, a ring of 36 traps, 100 random traps, a quasi-periodic set of 119, a $5 \times 5 \times 3$ lattice, a large array (32×32 and the disordered 819 set) and shaped fields; the grids are 128, 256, 512 pixels, the iteration budgets are 10, 30, 100 and the time budgets are 0.05, 1, 60 s (the last is the converged row). Every cell runs on 20 seeds and reports the median and the inter-quartile range of efficiency, uniformity, relative standard deviation, time and iterations, on the ideal device and on the C08 device (every hologram scored against its device). The Pareto fronts are stored with the cells.

The committed results file holds 63 rows, covering 1 of the 24 engine methods (M01); 62 method–geometry cells are recorded as skipped with a reason (a method that does not accept the target, such as a two-dimensional method on a three-dimensional lattice, or a method not registered in that build). The remaining cells are not yet measured, so this report makes no comparative statement from the benchmark and states

Table 5: Calibration case C12 (10 × 10 traps at eight Fourier units): efficiency e and uniformity u of the analytic and Gerchberg–Saxton baselines as baked, against the values published by Di Leonardo et al. for the same task.

Method	e (baked)	u (baked)	σ (%)	e (published)	u (published)
M01	0.328	0.000	268.79	0.29	0.01
M02	0.749	0.199	37.40	0.69	0.01
M03	0.010	0.077	54.25	0.01	0.58
M04	0.817	0.506	19.97	0.94	0.60
M05	0.818	0.738	11.81	0.93	0.79
M06	0.807	0.967	1.13	0.93	0.99
M07	0.799	0.971	1.41	n/a	n/a

Table 6: Baked results per case at the case budget: the method with the lowest relative standard deviation among those with at least half the best efficiency, and the phase-fixed WGS (M07) and plain GS (M04) rows where they ran. e efficiency, σ relative standard deviation of the trap intensities (%).

Case	Best	e	σ	M07 σ	M07 e	M04 σ
C01	B5	0.817	0.00	0.00	0.817	7.3
C02	B5	0.800	0.00	0.78	0.798	19.5
C03	M07	0.794	0.27	0.27	0.794	18.7
C04	shaped potential	n/a	n/a	n/a	n/a	n/a
C05	M09	0.699	0.01	n/a	n/a	22.4
C06	M07	0.704	0.00	0.00	0.704	7.3
C07	M09	0.638	0.00	0.00	0.793	18.7
C08	B1	0.648	6.71	14.76	0.675	21.6
C09	M14	0.805	0.08	7.27	0.453	15.0
C10	B2	0.800	0.00	0.01	0.798	n/a
C12	M06	0.807	1.13	1.41	0.799	20.0
C13	M07	0.723	0.10	0.10	0.723	11.2

the per-case baked scores and the unit measurements instead. Table 7 lists the cells present.

6 The learned models

Two networks were trained offline with PyTorch, exported to ONNX with a list of operators the browser runtime supports, and evaluated on held-out sets written before the final training with other seeds. A model is released only if it meets its criterion; otherwise the baseline answers.

cgh-net (M17). A U-Net maps three plane amplitudes, at −8, 0 and +8 SLM-limited Rayleigh ranges, to one phase for the 128 × 128 SLM, in the setting of DeepCGH [14]. It is trained without labels through the differentiable FFT model, with the loss $1 - \mathrm{AC}$ over the planes, $\mathrm{AC} = 2\sum TI/(\sum T^2 + \sum I^2)$ on unit-sum plane intensities, for 16000 steps. The exported graph is 4.67 MB. The baseline is volumetric three-dimensional GS (M19) on the same traps with a wall-clock budget of 7.35 s, the median time of M19 at 30 iterations. On 162 held-out volumes the network answers in about 35 ms with a median accuracy ratio of 1.140 (quartiles 1.102 and 1.160), above the baseline on 100 percent of the volumes; the median accuracy is 0.687 against 0.600. By kind the ratio is 1.162 (lattices), 1.147 (random), 1.118 (mixed), 1.100 (rings), 1.065 (the $5 \times 5 \times 3$ benchmark lattice) and 1.188 (the three-plane layout). The ONNX graph agrees with PyTorch on the reconstructed plane intensities to 8.4×10^{-6} (relative to the maximum). The accuracy measure scores full plane images, ghosts and blur included, which is what the network optimises; M19 optimises trap uniformity, so the ratio is a statement about this measure. The two-dimensional counterpart, a single-plane network followed by GS refinement, ties: on 219 held-out targets the ratio at equal wall-clock is 0.9994, below the criterion, and that model is not released.

warmstart-model (BS-3). A permutation-equivariant set network predicts initial trap phases for phase-fixed WGS. On 66 held-out geometries it needs a median of 8 iterations to reach 1 percent against 14 from a random start (a factor of 1.75, below the factor of two set as the criterion), but only with the early phase freeze (after 3 iterations, against 12 for the random arm). A control with no network, a random start and the same early freeze, needs a median of 9 iterations on the control subset, and 40 percent of its geometries never reach 1 percent (47 percent with the network); the final efficiency differs by 4.2 percent between the arms. The gain comes from the freeze, not from the network, so the proposal is refuted and the model is not released (Fig. 5).

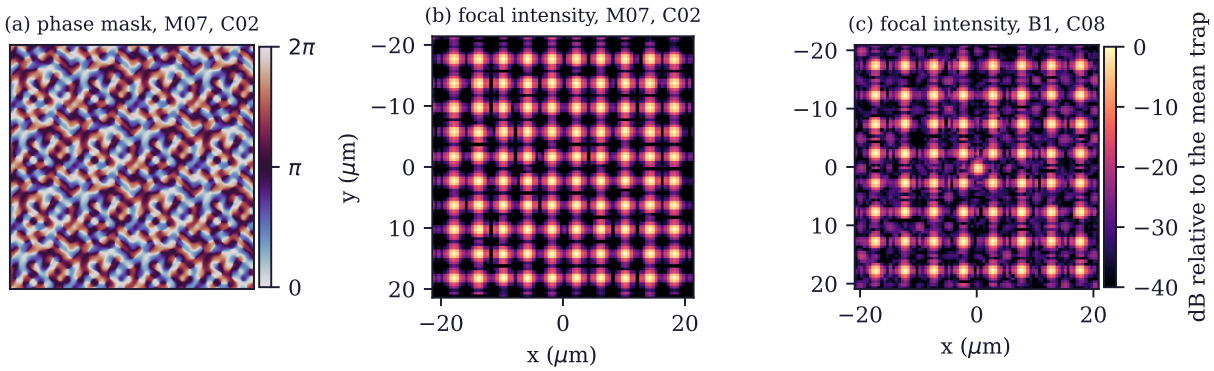


Figure 3: (a) Phase mask of the reference method on C02. (b) Its focal intensity, in dB relative to the mean trap, on the ideal device: a regular lattice. (c) The focal intensity of the device-aware weighted GS on C08, scored on the device with 4-bit phase, crosstalk and a fill factor of 0.85: the traps are visible and not equal.

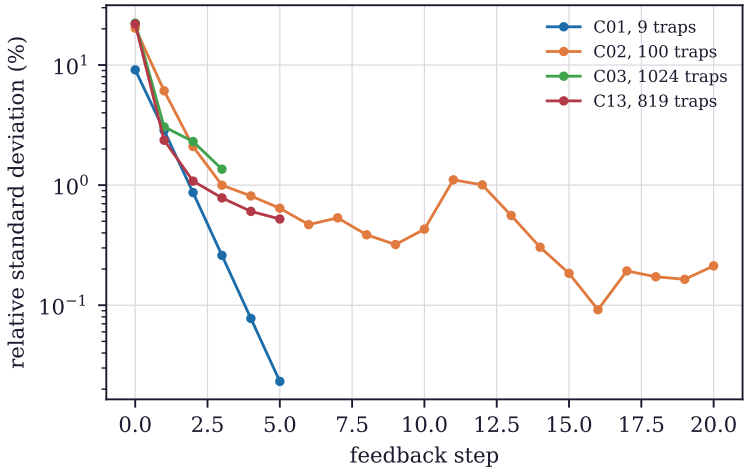


Figure 4: Emulated camera feedback: relative standard deviation of the trap intensities against the feedback step, on four cases (baked traces).

7 Proposals beyond the published state of the art

Each proposal is DinHot's own, grounded in a stated gap, measured against the strongest baseline and reported whether or not it succeeds.

BS-1, device-aware weighted GS. The weights of weighted GS are kept and the phase update is replaced by conjugate-gradient steps through the crosstalk kernel and the quantizer. The gap is that the crosstalk model has only been used with conjugate gradient on shaped potentials [9], or inside a weighted GS that replaces amplitudes only [17]. On the crosstalk device, the relative standard deviation is lower than that of phase-fixed WGS by the factor reported in the companion manuscript; on the small C08 grid the baked factor is 2.2 at an efficiency ratio of 0.960. **BS-2, flicker-bounded phase-continuous sequences.** Each intermediate hologram of a moving-trap sequence is optimized with the trap phases locked to a smooth trajectory; the gap is that linear phase interpolation does not optimize its intermediate frames [13]. The moving traps keep a higher minimum intensity than under linear phase interpolation, as quantified in the companion manuscript. **BS-3, learned warm start:** refuted (Section 6). **BS-4, aberration-robust hologram:** the expected cost over residual aberrations is minimized; over a 200-sample ensemble of residual Noll modes 4 to 11 the 95th percentile of the relative standard deviation is 1.55 times below that of M07 with an efficiency 1.9 percent lower (Table 3); robustness to pointing errors does not follow (1.02 times), because the spot width bounds it, and is reported as a negative result. The gap is that corrections are global or per site, never robust [18]. **BS-5, efficiency–uniformity Pareto front** by continuation of a weighted cost: the front's hypervolume exceeds that of the points of the GS family and one front point dominates M04 and M05 (Table 3), where published methods report single trade-off points [2].

Table 7: Benchmark cells measured so far in the committed results file (budget of 100 iterations, medians over the seeds). A cell absent from the file is not yet measured.

Method	Geometry	Device	Seeds	e	σ (%)
M01	G1	device	20	0.193	264.60
M01	G1	ideal	20	0.328	268.79
M01	G2	device	20	0.339	3.69
M01	G2	ideal	20	0.568	2.12
M01	G3	device	20	0.512	13.22
M01	G3	ideal	20	0.672	7.84
M01	G4	device	20	0.313	160.55
M01	G4	ideal	20	0.426	106.34
M01	G5	ideal	20	0.328	143.53

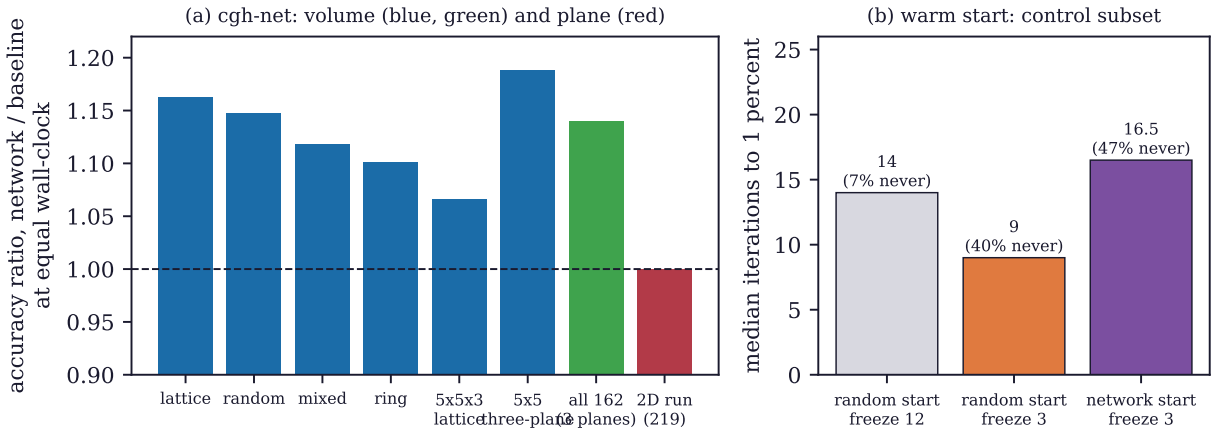


Figure 5: (a) The cgh-net accuracy ratio over its baseline at equal wall-clock, by target kind on the held-out volumes (G5: the $5 \times 5 \times 3$ lattice; 3-plane: the 5×5 three-plane layout), for the volume network (blue, green) and the two-dimensional single-plane network (red); the dashed line is the baseline. (b) The warm-start control: median iterations to 1 percent for a random start with the original and the early freeze and for the network start with the early freeze.

8 The live engine and the product

The site is static; every computation runs in the browser or was baked offline and is replayed. The live engine is a TypeScript port of the forward models, the device model and the live methods, running in a Web Worker so the interface does not block, with the FFT in Float64 (fft.js). An optional WebGPU backend runs the FFT-form iterations on the GPU and falls back to the CPU path when the browser offers none; an agreement check inside every GPU run compares the trap fields with the CPU result. At the live grid (256^2 padded to 512^2) a weighted GS iteration measured about 24 ms, so a 30-iteration run fits the lane budget of 1500 ms. The Python engine writes 46 reference documents and 104 unit tests compare the TypeScript engine with them; the worst observed difference of any quantity is 1.8×10^{-11} (the Parseval sum), and every phase agrees to 10^{-11} rad or better. On an integrated Intel gen-12lp adapter, 30 iterations of M07 took 0.30 to 0.32 s with the GPU backend against 1.5 to 1.6 s on the CPU path in the same browser. The methods that are baked and replayed are labelled as such in the interface. Of the 11 data contracts of the build, 6 are fully available; the others are partial (the case outside the field is refused, so one of the 13 case records is rejected by the input contract, and benchmark cells are not yet measured), answered by a baseline (the two model slots) or missing (the dormant API, which serves GET requests and answers 503 with the declared state).

9 Limits

DinHot computes and evaluates holograms in simulation: it does not drive an SLM or a camera, and its camera-in-the-loop method runs on an emulated camera whose errors are stated parameters. Device and set-up parameters come from published experiments, so absolute numbers belong to the stated model and not to a measured instrument. The engine is scalar (Fraunhofer and angular spectrum); the high-numerical-aperture

case carries a validity flag and the vectorial lane is not built. The fill-factor, 4-bit and flicker effects of the device have no published source in this build and are computed only. The crosstalk kernel's frequency unit is carried over from the engine and is not verified against the source. The transient-intensity model of BS-2 is the engine's own (Section 7 and the companion manuscript) and the linear-phase-interpolation baseline is DinHot's implementation of a published description, not the authors' code. The unit measurements of Table 3 use between one and six seeds; the benchmark that repeats them on 20 seeds is incomplete in the committed file. The cgh-net advantage is on a plane-image accuracy measure over generated volumes and the benchmark's volumetric layouts, not on a hardware experiment. The learned warm start is refuted and the two-dimensional network ties with GS. The executive figures of the introduction are taken from the cited publications and not re-measured.

Data and code availability

The source code, the baked artifacts (manifests, traces, benchmark results, model cards, the ONNX graph and its held-out metrics) and the scripts that regenerate every table and figure of this report from them are in the repository `https://github.com/fsantibanezleal/CEFOP_DinHot` (Apache-2.0), under `manuscripts/holographic-trap-studio/`; the studio runs at `https://dinhot.fasl-work.com`. Unit measurements quoted in Table 3 are produced by the tests named in Table 2.

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